

Recall:



rot + translation



$$\vec{r} \Rightarrow \vec{r}^*$$

rot.

$$\vec{r}^* = \vec{r}^*$$

$$r_i^* \hat{e}_i = r_i \hat{e}_i$$

to rotate,
keep components
but rotate bases

either Δ components
 Δ vectors

Rotation Matrix:

$$R = \left[\begin{array}{c|c|c} [\hat{e}_1']_f & [\hat{e}_2']_f & [\hat{e}_3']_f \end{array} \right]$$

each column is
components of one of $\hat{e}_i$
in f base vectors

$$R_{ij} = \hat{e}_i \cdot \hat{e}_j'$$

New base vectors are linear
combinations of old base
vectors

3 uses of $[R]$ (actually 6, because can use $[R]^T$ to "go the other way")

1) Represent rotation

$$\text{matrix: } [\vec{r}^*]_f = R \cdot [\vec{r}]_f$$

$$\text{components: } r_i^* = R_{ij} r_j$$

2) Find components of a given vector in one system from components in another system

$$\vec{r} = \vec{r}$$

prime: basic vector using i's prime

$$r_i \hat{e}_i = r'_i \hat{e}'_i$$

$$\hookrightarrow r_{\beta i} \hat{e}'_i = R_{\beta i} r'_i \hat{e}_i$$

$$\hat{e}_j \cdot \{ \} = r_i \hat{e}_j \cdot \hat{e}_i = r'_i \underbrace{\delta_{ji}}_{\delta_{ji}} \cdot \hat{e}'_i$$

$$\boxed{r_j = R_{ji} r'_i}$$

$$r_i = R_{ij} r'_j$$

$$[\vec{r}]_F = R [\vec{r}]_P$$

Similarly, $\{ \} \cdot \hat{e}'_j$

$$\Rightarrow [\vec{r}]_P = R^T [\vec{r}]_F$$

$$\Rightarrow \boxed{R^T = R^{-1}}$$

3) Change of basis

$$\hat{e}'_i = \underbrace{(\hat{e}'_i \cdot \hat{e}_j)}_{R_{ji}} \hat{e}_j$$

Similarly $\hat{e}'_i = R_{ij} \hat{e}_j$

Prads

$\vec{a} \vec{b}$ two vectors side by side

$$(\vec{a} \vec{b}) \cdot \vec{c} = \vec{a} (\vec{b} \cdot \vec{c})$$

Follows multiplication rules except
commutative

$$\vec{c} \cdot (\vec{a} \vec{b}) = (\vec{c} \cdot \vec{a}) \vec{b}$$